

# **OPERATING SYSTEMS — FULL PORTIONS**

## **UNIT 1 — INTRODUCTION TO OPERATING SYSTEMS**

### **1. Operating System Basics**

- Definition of Operating System
- Need for Operating System
- Goals of Operating System
- Functions of Operating System
- Evolution of Operating Systems

Types of Operating Systems

Batch OS

- Multiprogramming OS
- Multitasking OS
- Time-sharing OS
- Distributed OS
- Network OS
- Real-time OS
- Embedded OS
- Mobile OS

### **2. Computer System Organization**

- Computer system components
- CPU
- Main memory
- I/O devices
- Storage devices
- System bus

- Interrupts
- DMA

### **3. Operating System Structure**

- OS components
- Kernel
- User mode
- Kernel mode
- Dual-mode operation
- OS architecture
- Monolithic structure
- Layered structure
- Microkernel
- Modular structure
- Hybrid systems

### **4. Operating System Services**

- Program execution
- I/O operations
- File-system manipulation
- Communication
- Error detection
- Resource allocation
- Accounting
- Protection and security

### **5. User–OS Interface**

- Command Line Interface

- GUI
- Shell
- System programs

## **6. System Calls**

- System call concept
- System call interface

Types of system calls

Process control

- File management
- Device management
- Information maintenance
- Communication
- Protection

## **7. System Boot**

- Boot process
- BIOS/UEFI
- Bootloader
- Kernel loading
- OS initialization

## **8. Virtualization**

- Virtual machines
  - Hypervisors
  - Type 1 hypervisor
  - Type 2 hypervisor
  - Virtualization in OS
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# UNIT 2 — PROCESS MANAGEMENT

## 1. Process Concept

- Program vs Process
- Process states
- Process state diagram
- Process Control Block — PCB
- Process creation
- Process termination

## 2. Process Operations

- Parent and child processes
- Process creation
- `fork()`
- `exec()`
- `wait()`
- `exit()`
- Zombie process
- Orphan process

## 3. Process Scheduling

- Scheduling concept
- Scheduler
- Long-term scheduler
- Short-term scheduler
- Medium-term scheduler
- Dispatcher
- Context switching

## **4. Scheduling Criteria**

- CPU utilization
- Throughput
- Turnaround time
- Waiting time
- Response time

## **5. CPU Scheduling Algorithms**

- FCFS
- SJF
- SRTF
- Priority Scheduling
- Round Robin
- Multilevel Queue
- Multilevel Feedback Queue

## **Numerical Problems**

- Gantt chart
- Completion Time
- Turnaround Time
- Waiting Time
- Response Time
- Average waiting time
- Average turnaround time

## **6. Threads**

- Thread concept
- Process vs Thread

- Benefits of threads
- User-level threads
- Kernel-level threads
- Multithreading

Multithreading models

Many-to-One

- One-to-One
- Many-to-Many

## **7. Inter-Process Communication**

- IPC
- Shared memory
- Message passing
- Pipes
- Sockets
- Signals

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# **UNIT 3 — PROCESS SYNCHRONIZATION AND DEADLOCKS**

## **1. Process Synchronization**

- Race condition
- Critical section
- Critical-section problem
- Requirements of critical section

## **2. Synchronization Solutions**

- Peterson's solution

- Hardware synchronization
- Atomic operations
- Test-and-set
- Compare-and-swap

### 3. Semaphores

- Semaphore concept
- Binary semaphore
- Counting semaphore
- `wait()`
- `signal()`

### 4. Classical Synchronization Problems

- Producer–Consumer problem
- Readers–Writers problem
- Dining Philosophers problem

### 5. Monitors

- Monitor concept
- Condition variables
- Monitor-based synchronization

### 6. Deadlocks

- Deadlock concept

Necessary conditions

Mutual exclusion

- Hold and wait
- No preemption

- Circular wait

## **7. Deadlock Handling**

- Deadlock prevention
- Deadlock avoidance
- Deadlock detection
- Deadlock recovery

## **8. Deadlock Algorithms**

- Resource Allocation Graph
- Banker's Algorithm
- Safety algorithm
- Resource-request algorithm
- Deadlock detection algorithm

### **Important Numericals**

- Banker's Algorithm
- Safe sequence
- Resource allocation
- Deadlock detection

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## **UNIT 4 — MEMORY MANAGEMENT**

### **1. Memory Management Basics**

- Memory hierarchy
- Address binding
- Logical address
- Physical address



- MMU
- Dynamic loading
- Dynamic linking

## **2. Contiguous Memory Allocation**

- Single partition
- Multiple partitions
- Fixed partitioning
- Variable partitioning

## **3. Allocation Techniques**

- First Fit
- Best Fit
- Worst Fit
- Next Fit

## **4. Fragmentation**

- Internal fragmentation
- External fragmentation
- Compaction

## **5. Swapping**

- Swapping concept
- Swap space
- Swapping techniques

## **6. Paging**

- Paging concept

- Pages
- Frames
- Page table
- Address translation
- TLB
- Multilevel paging
- Inverted page table

## **7. Segmentation**

- Segmentation concept
- Segment table
- Logical address
- Segmentation with paging

## **8. Virtual Memory**

- Virtual memory concept
- Demand paging
- Page fault
- Page-fault handling
- Copy-on-write

## **9. Page Replacement Algorithms**

- FIFO
- Optimal
- LRU
- Second Chance
- Clock algorithm

## **Important Numericals**

- Page-table address calculation
- Page replacement
- Page faults
- FIFO
- Optimal
- LRU
- TLB
- Effective access time

## **10. Thrashing**

- Thrashing concept
- Causes
- Working-set model
- Page-fault frequency
- Thrashing control

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# **UNIT 5 — FILE SYSTEM AND I/O MANAGEMENT**

## **1. File System**

- File concept
- File attributes
- File operations
- File types

File access methods

Sequential

- Direct

- Indexed

## **2. Directory Structure**

- Single-level directory
- Two-level directory
- Tree-structured directory
- Acyclic graph directory
- General graph directory

## **3. File-System Implementation**

- File-system structure
- File-system mounting
- File control block
- In-memory file system

## **4. File Allocation Methods**

- Contiguous allocation
- Linked allocation
- Indexed allocation

## **5. Free-Space Management**

- Bit vector
- Linked list
- Grouping
- Counting

## **6. File Protection**

- Access control

- Access Control List — ACL
- Protection mechanisms

## **7. I/O Systems**

- I/O hardware
- I/O controller
- Device drivers
- Interrupt-driven I/O
- DMA
- Buffering
- Caching
- Spooling

## **8. Disk Management**

- Disk structure
- Disk scheduling
- Disk formatting
- Boot block
- Bad blocks

## **9. Disk Scheduling Algorithms**

- FCFS
- SSTF
- SCAN
- C-SCAN
- LOOK
- C-LOOK

## **Important Numericals**

- Disk head movement
- FCFS
- SSTF
- SCAN
- C-SCAN
- LOOK
- C-LOOK

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## **UNIT 6 — CASE STUDY / MODERN OS**

Depending on the Anna University regulation, this portion can appear as a separate unit or be incorporated into the other units.

### **Linux**

- Linux history
- Linux architecture
- Linux kernel
- Kernel modules
- Process management
- Linux scheduling
- Linux memory management
- Linux file system
- Linux I/O
- Linux IPC
- Linux networking
- Linux security

### **Windows**

- Windows architecture
- System components
- Process management
- Memory management
- File system
- Networking
- Security

## **Modern OS Concepts**

- Android OS
- Mobile operating systems
- Virtual machines
- Containers
- Cloud operating systems